

PART-1 : MATHEMATICS

SECTION-I (i)

1) Let

$$S_1 = \{(i, j, k) : i, j, k \in \{1, 2, \dots, 10\}\},$$

$$S_2 = \{(i, j) : 1 \leq i < j + 2 \leq 10, i, j \in \{1, 2, \dots, 10\}\}$$

$$S_3 = \{(i, j, k, \square) : 1 \leq i < j < k < \square, i, j, k, \square \in \{1, 2, \dots, 10\}\}$$

$$S_4 = \{(i, j, k, \square) : i, j, k \text{ and } \square \text{ are distinct elements in } \{1, 2, \dots, 10\}\}$$

If the total number of elements in the set S_r is n_r , $r = 1, 2, 3, 4$, then which of the following statements is(are) TRUE ?

(A) $n_1 = 1000$

(B) $n_2 = 44$

(C) $n_3 = 220$

(D) $\frac{n_4}{12} = 420$

2) Let α is satisfying the equation $\sin^{-1}(e^x) - \sin^{-1}(x^2) = 0$, $\alpha \in \mathbb{R}$ and β is number of solution of the

equation $\cos^{-1}\left(\frac{1 - \tan^2\left(\frac{x}{2}\right)}{1 + \tan^2\frac{x}{2}}\right) = |x - \pi| + \pi$, then which of the following is true ?

(A) $2\alpha + 3\beta > 0$

(B) $2\alpha + 3\beta < 0$

(C) $3\alpha + 2\beta > 0$

(D) $3\alpha + 2\beta < 0$

3) If $\sin^{-1}(\sin 10) - \tan^{-1}\tan(-6) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 4) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7) = p\pi - q$; where $p, q \in \mathbb{I}$, then which of the following is correct ?

(A) $3p - q = -4$

(B) $2p + q = 40$

(C) $p - 2q = 18$

(D) $q - 3p = 4$

4) Let $f_1(x) = x^2 + 4x + 2$ and for $n \geq 2$. Let $f_n(x)$ be the n -fold composition of the polynomial with itself. For example

$f_2(x) = f_1(f_1(x)) = x^4 + 8x^3 + 24x^2 + 32x + 14$ and S_n be the sum of the coefficients of the terms of even degree in $f_n(x)$. For example $S_2 = 1 + 24 + 14 = 39$. Then identify correct options.

(A) S_{2025} is divisible by 2

(B) S_{2025} is divisible by 3

(C) S_{2024} is divisible by 5

(D) S_{2024} is divisible by 3

5) Which of the following is/are correct ?

(A) If R and S are equivalence relations on a finite set A, then the composite relation R o S must also be an equivalence relation, where $n(A) \geq 3$.

(B) Let A be a set containing 5 elements. Then number of reflexive relations on A must be 2^{20}

(C) Let A be a set containing 5 elements. Then number of symmetric relations on A must be 2^{10}

(D) Let A be a set containing 4 elements. Then number of equivalence relations on A must be 15.

6) Let $\phi(x) = \tan \left(\frac{\cos^{-1} (\sin (\cos^{-1} x)) + \sin^{-1} (\cos (\sin^{-1} x))}{2} \right)$ then

(A) $\tan^{-1} \left(\phi \left(\frac{x}{2} \right) \right) = \frac{\pi}{4} \forall x \in [-2, 2]$

(B) $\sum_{x=1}^4 \phi \left(\frac{x}{4} \right) = 4$

(C) $\sin^{-1} (\phi (2x)) = \frac{\pi}{2} \forall x \in \left[-\frac{1}{2}, \frac{1}{2} \right]$

(D) $\sin^{-1} \left(\phi \left(\frac{x}{4} \right) \right) + \cos^{-1} \left(\phi \left(\frac{x}{3} \right) \right) = \frac{\pi}{4} \forall x \in [-3, 3]$

SECTION-I (ii)

1) Match the entries of List-I with one or more than one entries of List-II.

List-I		List-II	
(I)	Let $f : [-1, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt[5]{x} + \sin^{-1} x$ then $f(x)$ is	(P)	Odd
(II)	Let $f : \mathbb{R} \rightarrow \{-1, 0, 1\}$ is defined by $f(x) = \operatorname{sgn} \left(\frac{1 - x }{1 + x } \right)$ then $f(x)$ is ($\operatorname{sgn}(x)$ denotes signum function of x)	(Q)	Even
(III)	Let $f : (-\infty, 0] \rightarrow [0, \infty)$ be defined by $f(x) = \frac{2^{-[x]}}{2^{\{x\}}} - 2^{ x }$, then $f(x)$ is ($[x]$ denotes the largest integer less than or equal to x and $\{x\}$ denotes fractional part of x)	(R)	Onto
(IV)	Let $f : (-4, 2] \rightarrow [0, 3]$ be defined by $f(x) = \sqrt{8 - 2x - x^2}$, then $f(x)$ is	(S)	One-one
		(T)	Many-one

(A) $I \rightarrow P, S; II \rightarrow Q, R, T; III \rightarrow T; IV \rightarrow R, T$

(B) $I \rightarrow P, S; II \rightarrow R, T; III \rightarrow Q, R, T; IV \rightarrow T$

(C) $I \rightarrow P, S; II \rightarrow R, T; III \rightarrow T; IV \rightarrow Q, R, T$

(D) $I \rightarrow P, S; II \rightarrow T; III \rightarrow Q, R, T; IV \rightarrow R, T$

2) Match the following statement in List-I with the values in List-II

List-I		List-II	
(I)	If the domain of function $f(x) = \log_{ x-1 - x+1 - \frac{1}{2}} \left((x^2 - x + 1)^{x^2 - 3x + 2} - 1 \right)$ is the set D then the number of integer(s) which is(are) greater than -5 in D, is	(P)	0
(II)	Number of integer(s) in the range of the function $f(x) = \sqrt{x^2 + 4x} C_{2x^2 + 3}$, is	(Q)	2
(III)	If $f : \mathbb{R} \rightarrow \mathbb{R}$ and then $(f^{-1}(25) - f(25) + f(f(2)))$ is (where $[.]$ & $\{.\}$ denote greatest integer & fractional part function respectively)	(R)	4
(IV)	The number of solution(s) of the equation $2\tan^{-1}(\sec^2 \pi x) = \sin^{-1}(x^3 - x^2 + x + 2)$, is	(S)	6
		(T)	1

(A) $I \rightarrow R; II \rightarrow Q; III \rightarrow T; IV \rightarrow S$

(B) $I \rightarrow S; II \rightarrow T; III \rightarrow R; IV \rightarrow Q$

(C) $I \rightarrow R; II \rightarrow T; III \rightarrow Q; IV \rightarrow P$

(D) $I \rightarrow S; II \rightarrow Q; III \rightarrow T; IV \rightarrow P$

3) Match the entries of List-I with one or more than one entries of List-II.

List-I		List-II	
(I)	Let $f(n) = \sum_{k=-n}^n \left(\cot^{-1} \left(\frac{1}{k} \right) - \tan^{-1} k \right)$, $k \neq 0$. If the value $\sum_{n=2}^{10} (f(n) + f(n-1))$ is $m\pi$, then the value of $\left(\frac{m}{9} \right)$ is	(P)	1
(II)	If the sum of series $\sum_{p=1}^3 \sum_{r=1}^p \sum_{n=1}^{20} \sum_{m=1}^{20} \tan^{-1} \left(\frac{m}{n} \right) = \lambda$, then the sum of digits of $\frac{\lambda}{\pi}$ is	(Q)	4
(III)	If $[x]^3 + [y]^3 + [z]^3 - 3[x][y][z] = 4$ where $x, y, z > 0$, then minimum possible value of $(x + y + z)$ is (where $[.]$ denotes greatest integer function)	(R)	6
(IV)	Number of value(s) of x satisfying the equation $2\cos^{-1}(1 - x) - 3\cos^{-1}x = \pi$, is	(S)	9

		(T)	11
--	--	-----	----

- (A) $I \rightarrow T; II \rightarrow R; III \rightarrow S; IV \rightarrow Q$
 (B) $I \rightarrow S; II \rightarrow P; III \rightarrow T; IV \rightarrow Q$
 (C) $I \rightarrow T; II \rightarrow R; III \rightarrow Q; IV \rightarrow P$
 (D) $I \rightarrow S; II \rightarrow Q; III \rightarrow R; IV \rightarrow T$

4) Let $A = \{1, 2, 3, 4, \dots, 24, 25, 26\}$

A relation R is defined from set A to itself, then the number of

	List-I		List-II
(I)	Relation R if $R = \{(a, b) : R \text{ is asymmetric relation}\}$ are p^q (where p is prime number) then $(p + q)$ equals	(P)	328
(II)	Relations R if $R = \{(a, b) : R \text{ is reflexive but not symmetric}\}$ are $p^q(p^q - r)$, then $p + q + r$ is (where p, q, r $\in \mathbb{N}$ and p is prime number)	(Q)	329
(III)	Non-empty relations R if $R = \{(a, b) : R \text{ is symmetric but not reflexive}\}$ are $p^q(p^r - 1) - 1$ then $p + q + r$ is (where p is prime number)	(R)	330
(IV)	Non-empty relations R if $R = \{(a, b) : R \text{ is neither symmetric nor reflexive}\}$ are $(p^s - 1)p^q(p^q - 1)$ then $p + q + s$ is (where p is prime number)	(S)	353
		(T)	356

- (A) $I \rightarrow P; II \rightarrow P; III \rightarrow S; IV \rightarrow S$
 (B) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow S$
 (C) $I \rightarrow R; II \rightarrow Q; III \rightarrow T; IV \rightarrow T$
 (D) $I \rightarrow R; II \rightarrow P; III \rightarrow S; IV \rightarrow T$

SECTION-II

1) Find $f(\sqrt{2}) + f(-\sqrt{2}) + f(\pi) + f(-\pi)$, $f(a) = \left(a + \frac{(a+1)\sqrt{8a-1}}{\sqrt{27}}\right)^{\frac{1}{3}} + \left(a - \frac{(a+1)\sqrt{8a-1}}{\sqrt{27}}\right)^{\frac{1}{3}}$ if .

2) Consider a function $f(t) = \sin^{-1} \frac{2t}{1+t^2} + \cos^{-1} \frac{1-t^2}{1+t^2} + \tan^{-1} \frac{2t}{1-t^2}$; $t \neq \pm 1$. Then the value of $\frac{f\left(\frac{1+\sqrt{2}}{1-\sqrt{2}}\right)}{\tan^{-1}(3+2\sqrt{2})} + \frac{f\left(\frac{1}{\sqrt{2}}\right)}{\sin^{-1} \frac{1}{\sqrt{3}}} + \frac{f(\sqrt{7})}{\sec^{-1} \sqrt{8}}$ equals

3) The number of integral values that belongs to the exhaustive set of domain of the function

$$f(x) = \sqrt{(2x+9) - \left(\frac{2x}{1-\sqrt{1+2x}}\right)^2} \text{ is}$$

4) If $\alpha = \sum_{n=1}^{\infty} \tan^{-1} \left\{ \frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16} \right\}$ and $\sin \alpha = \frac{p}{q}$ (where p, q are relatively prime numbers) then the value of $(p^2 + q^3)$ is

5) Let the function f and g be defined by

$$f(x) = \begin{cases} \sqrt{x} & \text{for } 0 \leq x < 1 \\ 2-x & \text{for } 1 \leq x < 2 \\ f(x+2) & \text{for all } x \end{cases} \text{ and } g(x) = 4f(3x) + 1 \text{ for all } x.$$

Let M denotes the sum of all the solutions of the equation $f(x) = 0.6$ for $3 \leq x \leq 7$

N denotes the fundamental period of g (x).

P denotes the value of $g'(6.75)$.

Then the value of $[M]NP$ is (where $[.]$ denotes the greatest integer function and $g'(x)$ denotes derivative of g(x) with respect to x)

6) All the distinct integers of the sequence $\left\lfloor \frac{1^2}{15} \right\rfloor, \left\lfloor \frac{2^2}{15} \right\rfloor, \left\lfloor \frac{3^2}{15} \right\rfloor, \dots, \left\lfloor \frac{15^2}{15} \right\rfloor$ constitute a set A. Then, number of injective functions in $A \rightarrow A$ such that

$$f(i) = \begin{cases} i+1; & i \leq 5 \\ i+2; & 9 \leq i \leq 13 \end{cases}$$

$f(i) \neq i$; otherwise

(where $i \in A$ and $[*]$ denotes greatest integer function)

7) The number of integral value(s) that does not belong to the exhaustive set of range of α such that

range of $f(x) = \frac{(x^3 - 3x + 6)(x - \alpha)}{2(x - \alpha)}$, for permissible value of x, is a proper subset of R, is

8) Let $f(x) = \frac{1}{\pi} (\sin^{-1} x + \cos^{-1} x + \tan^{-1} x) + \frac{x+1}{x^2 + 2x + 10}$. If the absolute maximum value of f(x) is M then the value of $(156M - 1)$ is

PART-2 : PHYSICS

SECTION-I (i)

1) A point object moves along the principal axis of a convex spherical mirror of focal length 20 cm. Initially, the object is 60 cm from the pole and moves toward the mirror with a constant speed of 8 cm s⁻¹. Consider the motion until the object reaches 20 cm from the pole. Which of the following

statements are correct?

- (A) During this interval, the image moves 5 cm toward the pole, while its speed increases continuously from 0.5 cm s^{-1} to 3 cm s^{-1} .
- (B) When the object is 20 cm from the pole, the magnitude of acceleration of the image is 0.8 cm s^{-2} , directed toward the pole.
- (C) At the instant when the image speed is 1.28 cm s^{-1} , the object is 30 cm from the pole and the magnitude of linear magnification is 0.4.
- (D) The average speed of the image during the complete motion is 1.25 cm s^{-1} .

2) A thin symmetric biconvex lens L_1 of refractive index 1.50 has radii of curvature $R_1 = +20 \text{ cm}$, $R_2 = -20 \text{ cm}$. A thin symmetric biconcave lens L_2 of refractive index 1.60 has radii $R_1 = -40 \text{ cm}$, $R_2 = +40 \text{ cm}$. The lenses are kept in contact and the combination is immersed completely in a homogeneous liquid of refractive index μ . Neglect the separation between the lenses and assume paraxial approximation. Which of the following statements are correct?

- (A) In air, the combination behaves as a converging lens of focal length 50 cm.
- (B) For $\mu = 1.40$, the equivalent focal length of the combination becomes infinite, although neither lens individually has infinite focal length.
- (C) For $\mu = 1.55$, both L_1 and L_2 behave as diverging lenses, and the equivalent focal length of the combination is $-\frac{31}{15} \text{ m}$.
- (D) For $\mu = 1.75$, L_1 behaves as a diverging lens and L_2 as a converging lens, but their combination is still converging.

3) Four modified vernier calipers have fewer vernier divisions than the corresponding number of main-scale divisions occupying the same length. For each instrument, 1 MSD = 1 mm. Which of the following statements are correct?

- (A) For 20 VSD = 25 MSD, the vernier zero lies between 12 mm and 13 mm and the 2nd VSD coincides with an MSD. If the zero error is +0.50 mm, the corrected reading is 12.00 mm.
- (B) For 15 VSD = 18 MSD, the vernier zero lies between 7 mm and 8 mm and the 2nd VSD coincides with an MSD. If the zero error is -0.20 mm, the corrected reading is 7.40 mm.
- (C) For 10 VSD = 14 MSD, the vernier zero lies between 18 mm and 19 mm and the 3rd VSD coincides with an MSD. If the zero error is +0.40 mm, the corrected reading is 18.40 mm.
- (D) For 8 VSD = 12 MSD, the vernier zero lies between 21 mm and 22 mm and the 3rd VSD coincides with an MSD. If the zero error is +0.50 mm, the corrected reading is 21.50 mm.

4) In a Searle's apparatus experiment, Young's modulus of a wire is determined using $Y = \frac{4MgL}{\pi d^2 \Delta \ell}$. Take g and π as exact. The positions of the two ends of the wire are $x_1 = (15.0 \pm 0.1) \text{ cm}$, $x_2 = (215.0 \pm 0.1) \text{ cm}$. The corrected screw-gauge readings for the diameter are

Reading	1	2	3	4	5
d(mm)	0.498	0.500	0.502	0.500	0.500

The Searle micrometer readings are $z_1 = (3.200 \pm 0.005) \text{ mm}$, $z_2 = (4.800 \pm 0.005) \text{ mm}$ and the applied mass is $M = (5.00 \pm 0.01) \text{ kg}$. Which of the following statements are correct ?

From the given end-position readings, screw-gauge observations, and Searle micrometer

- (A) readings, the experimentally determined quantities to be used in calculating Y are $L = (200.0 \pm 0.2) \text{ cm}$, $d = (0.500 \pm 0.002) \text{ mm}$, $\Delta l = (1.600 \pm 0.020) \text{ mm}$.
- (B) The percentage contributions to the maximum error in Y due to M , L , d , Δl are respectively 0.20%, 0.10%, 0.80%, 0.625%.
- (C) The maximum percentage error in Y is 1.725%.
- (D) The maximum percentage error in Y is 1.525%.

5) A spherical refracting surface of radius of curvature 20 cm separates air ($n_1 = 1$) on the left from glass ($n_2 = 1.5$) on the right. The centre of curvature lies in the glass. A point object is placed on the principal axis in air. Cartesian sign convention is used. Which of the following statements are correct?

- (A) If the object is 30 cm from the pole, the image is virtual and located 180 cm from the pole on the object side, with transverse magnification +4.
- (B) If the object is moved to 40 cm from the pole, the refracted rays emerge parallel to the principal axis.
- (C) If the object is 60 cm from the pole, a real image is formed 180 cm inside the glass.
- (D) As the object is moved continuously from 30 cm to 60 cm away from the pole, the image changes continuously from virtual to real by passing through the pole.

6) A student measures the dimensions and mass of a rectangular solid as $\ell = 12.40 \text{ cm}$, $b = 3.2 \text{ cm}$, $h = 0.850 \text{ cm}$, $m = 26.75 \text{ g}$. The student evaluates $V = \ell bh$, $\rho = \frac{m}{\ell bh}$, $S = \ell + b + h$, $Q = \frac{m\ell}{bh}$. Using standard rules of significant digits, which of the following statements are correct?

- (A) The volume should be reported as $V = 34 \text{ cm}^3$.
- (B) The density should be reported as $\rho = 0.79 \text{ g cm}^{-3}$.
- (C) The sum should be reported as $S = 16.45 \text{ cm}$.
- (D) The quantity Q should be reported as $Q = 1.2 \times 10^2 \text{ g cm}^{-1}$.

SECTION-I (ii)

1) Two plane mirrors M_1 and M_2 , with their reflecting surfaces facing each other, are inclined at an angle θ . A point object O is placed between them. Ignore the finite size of the mirrors. Match the configurations in List-I with the number of distinct images formed in List-II.

	List-I (Configurations)		List-II (number of distinct images)
(I)	$\theta = 60^\circ$; object placed at any point between the mirrors	(P)	4
(II)	$\theta = 72^\circ$; object lies on the angle bisector	(Q)	5
(III)	$\theta = 72^\circ$; object does not lie on the angle bisector	(R)	6

(IV)	$\theta = 50^\circ$; object placed at a general point between the mirrors	(S)	7
		(T)	8

- (A) $I \rightarrow Q; II \rightarrow P; III \rightarrow Q; IV \rightarrow S$
 (B) $I \rightarrow P; II \rightarrow Q; III \rightarrow Q; IV \rightarrow S$
 (C) $I \rightarrow P; II \rightarrow P; III \rightarrow R; IV \rightarrow S$
 (D) $I \rightarrow Q; II \rightarrow P; III \rightarrow Q; IV \rightarrow T$

2) A screw gauge is used to measure the diameter of different wires. For each observation in List-I, determine the corrected diameter and match it with List-II. Assume there is no backlash error.

	List-I (Observation)		List-II (Corrected diameter)
(I)	Pitch = 0.50 mm, 50 circular-scale divisions; zero error = +0.03 mm; MSR = 6.50 mm, CSR = 23	(P)	3.64 mm
(II)	Pitch = 1.00 mm, 100 circular-scale divisions; zero error = - 0.04 mm; MSR = 4.00 mm, CSR = 37	(Q)	7.715 mm
(III)	Pitch = 0.50 mm, 100 circular-scale divisions; zero error = +0.015 mm; MSR = 7.50 mm, CSR = 46	(R)	6.70 mm
(IV)	Pitch = 1.00 mm, 50 circular-scale divisions; zero error = - 0.06 mm; MSR = 3.00 mm, CSR = 29	(S)	4.41 mm
		(T)	7.73 mm

- (A) $I \rightarrow R; II \rightarrow Q; III \rightarrow S; IV \rightarrow P$
 (B) $I \rightarrow Q; II \rightarrow S; III \rightarrow R; IV \rightarrow P$
 (C) $I \rightarrow R; II \rightarrow S; III \rightarrow Q; IV \rightarrow P$
 (D) $I \rightarrow P; II \rightarrow S; III \rightarrow Q; IV \rightarrow R$

3) A plane interface perpendicular to the x-axis separates two transparent media. The interface itself may move along the x-axis with velocity V . An object moves normally to the interface with velocity v_0 . All velocities are measured in the laboratory frame; +x is towards the right. For each case in List-I, the object is viewed paraxially from the medium on the opposite side of the interface. Match each case with the velocity of its refracted image given in List-II.

	List-I (Configuration)		List-II (Velocity of refracted image)
(I)	$n_L = \frac{3}{2}$, $n_R = 1$; object in left medium, $V = +2$ m/s, $v_0 = +5$ m/s	(P)	4 m/s towards +x

(II)	$n_L = 1, n_R = \frac{4}{3}$; object in left medium, $V = +2$ m/s, $v_0 = -1$ m/s	(Q)	2 m/s towards -x
(III)	$n_L = 1, n_R = \frac{5}{3}$; object in right medium, $V = -1$ m/s, $v_0 = +4$ m/s	(R)	2 m/s towards +x
(IV)	$n_L = \frac{3}{2}, n_R = \frac{4}{3}$; object in right medium, $V = +1$ m/s, $v_0 = -5$ m/s	(S)	$\frac{23}{4}$ m/s towards -x
		(T)	6 m/s towards +x

- (A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ S
 (B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ T
 (C) I $\rightarrow$ Q; II $\rightarrow$ T; III $\rightarrow$ R; IV $\rightarrow$ S
 (D) I $\rightarrow$ Q; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ S

4) A thin lens is placed immediately in front of a spherical mirror, with negligible separation between them. Light is incident from the left, passes through the lens, reflects from the mirror, and passes through the lens again. All lenses are in air and paraxial approximation is valid. For the lens, use the Cartesian sign convention for R_1 and R_2 . The equivalent focal length F of the complete double-pass optical system is taken positive for a converging combination and negative for a diverging combination. Match each optical system in List I with its equivalent focal length in List II.

	List-I (Optical System)		List-II (Equivalent focal length F)
(I)	A glass lens of refractive index $\mu = \frac{3}{2}$, has $R_1 = +20$ cm, $R_2 = -20$ cm. It is in contact with a concave mirror of radius of curvature 60 cm.	(P)	∞
(II)	A glass lens of refractive index $\mu = \frac{3}{2}$, has $R_1 = -30$ cm, $R_2 = +30$ cm. It is in contact with a concave mirror of radius of curvature 40 cm.	(Q)	-40 cm
(III)	A glass lens of refractive index $\mu = \frac{3}{2}$, has $R_1 = +40$ cm, $R_2 = -40$ cm. It is in contact with a convex mirror of radius of curvature 40 cm.	(R)	$+\frac{40}{3}$ cm
(IV)	A glass meniscus lens of refractive index $\mu = \frac{3}{2}$ has $R_1 = +20$ cm, $R_2 = +40$ cm. It is in contact with a concave mirror of radius of curvature 40 cm.	(S)	$+\frac{15}{2}$ cm
		(T)	-60 cm

- (A) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ P; IV $\rightarrow$ R
 (B) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ P

(C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T

(D) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ R; IV $\rightarrow$ P

SECTION-II

1) In a simple pendulum experiment, the distance from the point of suspension to the top of the spherical bob is measured as $l = (78.0 \pm 0.1)$ cm, and the diameter of the bob is $d = (4.0 \pm 0.2)$ cm. The effective length of the pendulum is measured from the point of suspension to the centre of the bob. The times recorded for 25 complete oscillations are 39.7 s, 40.0 s, 40.3 s. For these repeated observations, take the random uncertainty in the measured total time to be the mean absolute error about the mean. In addition, the experiment has a maximum reaction-time uncertainty of 0.10 s at both the start and stop of each timing measurement. Neglect all other errors. If the maximum percentage uncertainty in the calculated value of acceleration due to gravity is $x\%$, find x .

2) A travelling microscope is used to determine the refractive index μ of a transparent glass slab. The scale reading increases upward. The readings are $R_1 = 7.844$ cm for the upper surface of the slab, $R_2 = 6.642$ cm for a fine mark on the table as seen through the slab and $R_3 = 6.044$ cm for the same mark viewed directly after removing the slab. The least count of the travelling microscope is 0.002 cm. Calculate the maximum percentage uncertainty in μ .

3) The surface tension S of a liquid is determined by the capillary-rise method using $S = \frac{\rho g h d}{4 \cos \theta}$, where d is the internal diameter of the capillary tube. The diameter is measured at five different orientations using a screw gauge, giving 0.82, 0.84, 0.83, 0.81, 0.85 mm. The screw gauge has a known positive zero error of +0.02 mm. Its least count is 0.01 mm. Take the uncertainty in the mean diameter as the larger of the least count and the mean absolute deviation of the five readings from their mean. Assume the zero correction is known exactly. Other measurements are $h = (5.00 \pm 0.02)$ cm, $\rho = (1000 \pm 5)$ kg m⁻³, $\theta = (30.0 \pm 0.5)^\circ$. Take g as exact. Use differential error propagation for the angular factor and express angular uncertainty in radians. If the maximum percentage uncertainty in the calculated surface tension is $x\%$, find x .

4) In an optical bench experiment, an object O, a thin convex lens L and a screen S are adjusted to obtain a sharp real image. Their bench readings are $x_o = 15.0$ cm, $x_L = 45.0$ cm, $x_s = 90.0$ cm. Each bench reading has a maximum possible error of ± 0.10 cm. Calculate the maximum percentage uncertainty in the measured focal length.

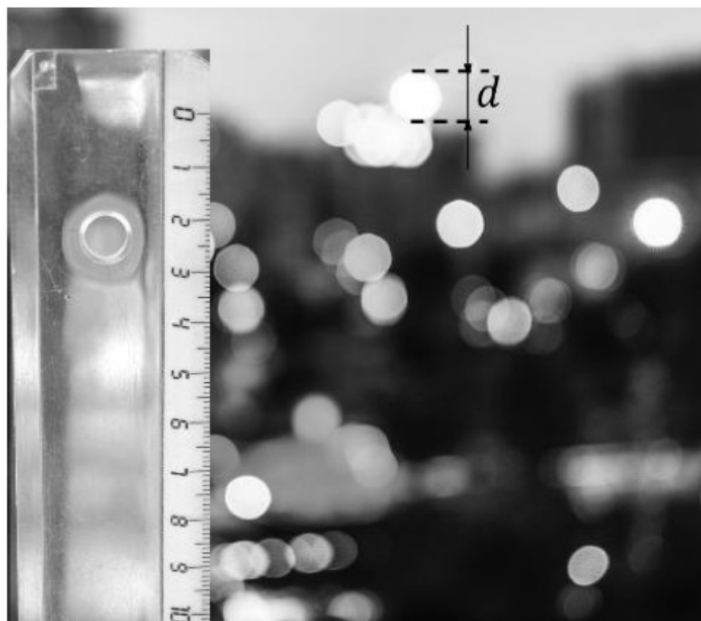
5) A potentiometer wire is used to determine the internal resistance r of a cell. When the cell is on open circuit, the balancing length is measured as $l_1 = (80.0 \pm 0.2)$ cm. When the cell is connected across an external resistance $R = (5.00 \pm 0.05)\Omega$, the balancing length becomes $l_2 = (64.0 \pm 0.2)$ cm. Assume the potentiometer current remains unchanged during the two measurements and neglect all other sources of error. Determine the maximum percentage uncertainty in the calculated internal resistance of the cell.

6) In a lens displacement experiment, an object and a screen are fixed on an optical bench. Their indicated positions are $x_o = 12.6$ cm, $x_s = 131.6$ cm. The object holder has a correction of -0.4 cm, while the screen holder has a correction of +0.6 cm. Thus, the true positions are obtained by adding the respective corrections to the indicated readings. A convex lens forms sharp images on the screen

at two indicated lens positions $x_1 = 48.3$ cm, $x_2 = 96.3$ cm. The lens holder has a constant index correction of $+0.3$ cm in both positions. Using the lens displacement method, determine the focal length (in cm) of the convex lens. Account correctly for all index corrections.

7) In a calorimetry experiment, a hot solid is dropped into water contained in a calorimeter. The calorimeter and the water are initially in thermal equilibrium. The heat capacity of the calorimeter is known exactly, and heat loss to the surroundings is negligible. The measured temperatures are $T_s = (90.0 \pm 0.1)^\circ\text{C}$, $T_w = (20.0 \pm 0.1)^\circ\text{C}$; where T_s is the initial temperature of the solid and T_w is the common initial temperature of the water and the calorimeter. The final equilibrium temperature is $T_f = (30.0 \pm 0.1)^\circ\text{C}$. The masses of the solid and water and the specific heat capacity of water may be regarded as exact. Using the principle of calorimetry, determine the maximum percentage uncertainty in the calculated specific heat capacity of the solid due only to the uncertainties in the three temperature measurements.

8) A camera is focused sharply on a transparent ruler placed perpendicular to its optical axis at a distance of 25.0 cm from a thin camera lens. A 12.0 cm length of the ruler forms a 3.00 cm long image on the camera sensor. In the same photograph, a distant point source appears as a circular blurred spot of diameter 8.40 mm. Assume that rays from the distant source are parallel to the optical axis and that the full circular aperture of the lens is used. Neglect diffraction and aberrations. Determine the diameter D of the camera lens, in cm.



PART-3 : CHEMISTRY

SECTION-I (i)

1) Select **CORRECT** about test of M^{+2} in V^{th} group cation analysis when M^{+2} produce brick red colour to flame.

- (A) It produce white precipitate with $(\text{NH}_4)_2\text{CO}_3$ in the presence of NH_4Cl and NH_4OH
- (B) It produce yellow precipitate with $(\text{NH}_4)_2\text{CrO}_4$ in the presence of $\text{CH}_3\text{COOH}_{(\text{aq.})}$

(C) Flame test looks greenish-yellow through blue glass.

(D) It produce white precipitate with $(\text{NH}_4)_2\text{C}_2\text{O}_4$ in the presence of CH_3COOH solution.

2) Select the option(s) where both the oxoanion of first transition series, in which metal exhibits the oxidation state equal to its group number in periodic table.

(A) VO_4^{3-} , CrO_4^{2-}

(B) FeO_4^{2-} , MnO_4^-

(C) MnO_4^{2-} ,
 VO_4^{3-}

(D) $\text{Cr}_2\text{O}_7^{2-}$, MnO_4^-

3) Which of the following group of ions can coexist in significant quantities in aqueous solution, without formation of any ppt. and showing redox reaction with each other?

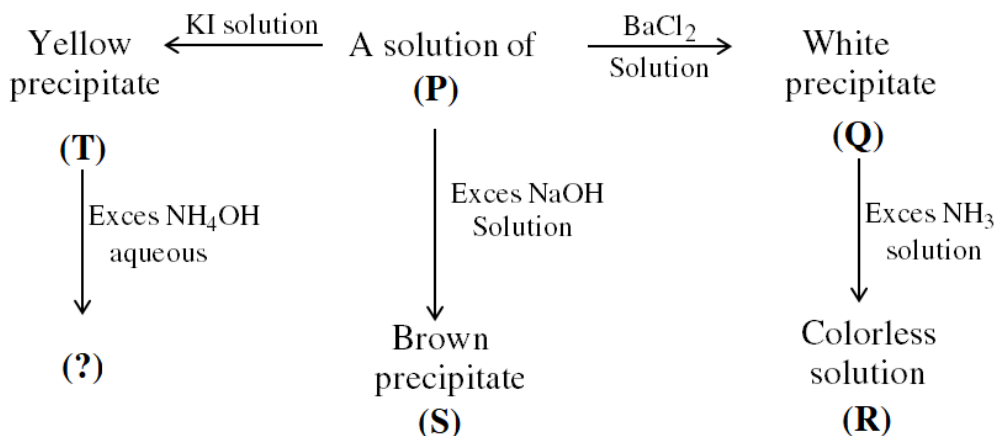
(A) Mn^{+2} , SO_4^{2-} , Al^{+3} , BiO_3^-

(B) Ba^{+2} , Sr^{+2} , S^{2-} , Cl^-

(C) K^+ , CO_3^{2-} , Cl^- , Al^{+3}

(D) Cu^{+2} , Ni^{2+} , Cl^- , SO_4^{2-}

4) Aqueous solution of **(P)** react in the following ways :



Select **CORRECT** option(s) for above reactions.

(A) **(P)** = PbSO_4 , **(Q)** = BaSO_4 , **(T)** = PbI_2

(B) **(P)** = AgNO_3 , **(R)** = $[\text{Ag}(\text{NH}_3)_2]^+\text{Cl}^-$, **(S)** = Ag_2O

(C) **(Q)** = AgCl , **(?)** = T , **(T)** = AgI

(D) **(P)** = $\text{Fe}_2(\text{SO}_4)_3$, **(S)** = $\text{Fe}(\text{OH})_3$, **(T)** = KI_3

5) Metal "**M**" of 3d-series which is a very abundant (ninth of all elements and second of the transition elements). The two most important minerals of Metal "**M**" are ilmenite and rutile. Dioxide of metal **M** is widely used as white pigment.

Select **CORRECT** option(s) about Metal **M** & its compounds -

- (A) Its Melting point is lower than Metal which placed just below it in periodic table but higher than element which is just next to it in same period.
- (B) Its +4 ion is also stabilise as oxocation MO^{2+}
- (C) It forms oxide and all halides (except iodide) in its highest oxidation state.
- (D) Aqueous solution of M(IV) ion is reduced by Zn and purple aqua ion $[\text{M}(\text{H}_2\text{O})_6]^{3+}$ is obtained.
- 6) A^{2+} , B^{2+} , C^{2+} and D^{2+} belongs to Group IV cationic analysis. If sulphide of A^{2+} is buff pink in colour, white hydroxide of B^{2+} dissolved in excess NH_3 solution. Aquated C^{2+} is pink and aquated D^{2+} is green in colour.
- Select the **INCORRECT** option(s).

- (A) All octahedral complexes of A^{2+} , C^{2+} and D^{2+} are paramagnetic in nature.
- (B) B^{2+} with excess NaOH solution gives coloured diamagnetic complex species.
- (C) Metal "**D**" is used as a catalyst during conversion of 2-ethyl anthraquinol to 2-ethyl anthraquinone in the production of H_2O_2 .
- (D) Conversion A^{+2} to AO_4^- can be done by heating with PbO_2 in acidic medium.

SECTION-I (ii)

1)

List-I (Complexes of 3d, 4d and 5d transition metals)		List-II (Electronic, bonding, Magnetic and structural characteristic)	
(P)	$[\text{Rh}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$	(1)	Low spin, diamagnetic and d^6 configuration of metal ion
(Q)	$[\text{Ir}(\text{Cl})(\text{CO})(\text{PPh}_3)_2]$	(2)	Can exhibits Linkage or Ionization or Geometrical isomerism
(R)	$[\text{Ru}(\text{H}_2\text{O})_6]\text{Cl}_3$	(3)	Square planar with dsp^2 hybridisation of metal ion
(S)	$[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$	(4)	Primary valency = 3 Secondary valency = 6 (According to Werner's theory)
		(5)	$(\mu_{\text{spin only}}) \neq 0$ with $\text{CFSE} = -2.0 \Delta_o$

Select correct code :-

- (A) $\text{P} \rightarrow 1,2; \text{Q} \rightarrow 1,3; \text{R} \rightarrow 4,5; \text{S} \rightarrow 2$
- (B) $\text{P} \rightarrow 1,2; \text{Q} \rightarrow 2; \text{R} \rightarrow 5; \text{S} \rightarrow 2,4$
- (C) $\text{P} \rightarrow 1,4; \text{Q} \rightarrow 2,3; \text{R} \rightarrow 1,4; \text{S} \rightarrow 2,4,5$
- (D) $\text{P} \rightarrow 4,5; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 4$

2) Consider the example of π -acid ligands given in List-II and their donor ability during complex formation in List-I find **CORRECT** code of matching :-

List-I (Type of π -acid ligand)		List-II (Example)	
(P)	Anionic $6e^-$ donor	(1)	Nitrosonium
(Q)	Cationic $2e^-$ donor	(2)	Triethylphosphine
(R)	Neutral $2e^-$ donor	(3)	Methylcyanide
(S)	Anionic $2e^-$ donor	(4)	Cyanide
		(5)	Cyclopentadienyl

Select **CORRECT** option is :-

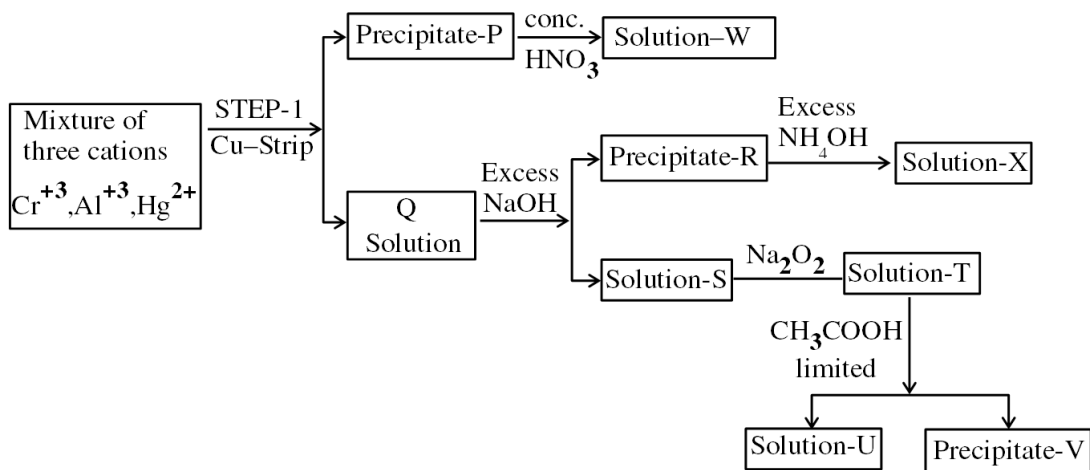
- (A) $P \rightarrow 5; Q \rightarrow 1,3; R \rightarrow 2; S \rightarrow 4$
 (B) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 2,3; S \rightarrow 4$
 (C) $P \rightarrow 3,5; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$
 (D) $P \rightarrow 4; Q \rightarrow 1,2; R \rightarrow 3; S \rightarrow 5$

3) Consider the product of reactions given in List-II and specification about product in reactions in List-I find **CORRECT** code of matching :-

List-I (Specification about product of reaction)		List-II (Reactions)	
(P)	Green coloured residue or solution	(1)	$K_2Cr_2O_7 \xrightarrow{H_2SO_3}$
(Q)	Paramagnetic (any product)	(2)	$Pyrolusite \xrightarrow[\Delta, \text{ fused}]{KOH+O_2}$
(R)	Tetrahedral geometry at central metal	(3)	$CuSO_{4(aq.)} \xrightarrow[HCl]{conc.}$
(S)	Redox change	(4)	$K_2CrO_{4(s)} \xrightarrow[HNO_3]{conc.}$
		(5)	$K_4[Fe(CN)_6] \xrightarrow[H_2O_2]{H_2SO_4}$

The **CORRECT** option is :

- (A) $P \rightarrow 1,2,5; Q \rightarrow 1,2,3,5; R \rightarrow 1,2,3,4; S \rightarrow 2,3,4$
 (B) $P \rightarrow 1,2; Q \rightarrow 1,2,4; R \rightarrow 3,4,5; S \rightarrow 2,3,5$
 (C) $P \rightarrow 1,2,3; Q \rightarrow 1,2,3,5; R \rightarrow 2,3,4; S \rightarrow 1,2,5$
 (D) $P \rightarrow 2,3; Q \rightarrow 1,2,3,4,5; R \rightarrow 2,3,4; S \rightarrow 1,2,3$



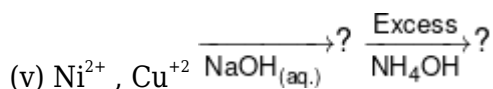
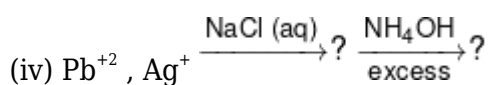
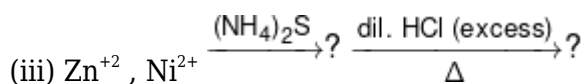
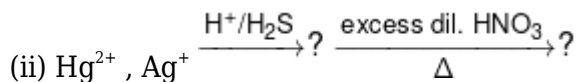
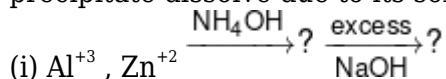
4)

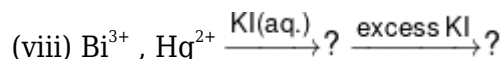
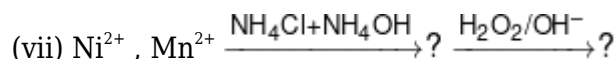
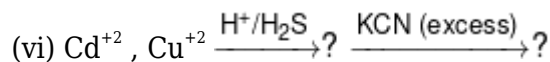
List-I (Colour of Solution)		List-II (Solutions in above reaction)	
(P)	Yellow / Orange	(1)	(W)
(Q)	Green	(2)	(S)
(R)	Blue / Dark blue	(3)	(X)
(S)	Colourless	(4)	(T)
		(5)	(U)

- (A) $\text{P} \rightarrow 4; \text{Q} \rightarrow 5; \text{R} \rightarrow 3; \text{S} \rightarrow 1$
- (B) $\text{P} \rightarrow 5; \text{Q} \rightarrow 2; \text{R} \rightarrow 3; \text{S} \rightarrow 1$
- (C) $\text{P} \rightarrow 4, 5; \text{Q} \rightarrow 5; \text{R} \rightarrow 2; \text{S} \rightarrow 3$
- (D) $\text{P} \rightarrow 1; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 5$

SECTION-II

1) Among the following reactions, Count the number of reactions in which both cations initially precipitated in Step-1, but the total volume of precipitate decreases in Step-2 because one of the precipitate dissolve due to its solubility in Step-2 reagent.





2) Find the value of x, y and z in the formulae of the following complexes by determining the oxidation state of Metal from the given μ_{spin} values of complexes :

(a) $[\text{VCl}_x(\text{bipy})]$, $\mu = 1.73$ B.M.

(b) $\text{K}_y[\text{V(ox)}_3]$, $\mu = 2.82$ B.M.

(c) $[\text{Mn(CN)}_6]^{z-}$ $\mu = 3.88$ B.M.

If t = total possible optically active isomers of complexes a, b and c

Write the value of $(x + y + z) \times t$

3) UK "silver" coins are not pure silver — they are made of an alloy of two metals of 3d series **M₁** & **M₂** only (no silver in alloy). 5.00 gram of UK "silver" coin dissolved completely in conc. H_2SO_4 and resulting coloured solution is diluted by water to produces **solution (S)**. H_2S gas is passed through the diluted **solution (S)** and **M₁^{+x}** ions are completely precipitated from solution as a black precipitate. The precipitate filtered & dried and weight = 5.64 gram.

The filtrate containing **M₂^{+x}** ion is now treated with $\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$, no precipitate is form but on passing H_2S in the presence of NH_4Cl & NH_4OH produces a black sulphide precipitate of **M₂^{+x}**. The precipitate is filtered & dried and weighted 1.93 gm.

Given :

M₁^{+x}(aq.) = Produce brown precipitate with $\text{K}_4[\text{Fe(CN)}_6]$

M₂^{+x}(aq.) = Produce blue hexaammine complex with excess NH_4OH

Calculate total mass (in gm) of hydroxide precipitate is formed if entire **solution (S)** react with excess NaOH .

(Atomic mass of Sc = 45, Ti = 48, V = 51, Cr = 52, Mn = 55, Fe = 56, Co = 59, Ni = 58.7, Cu = 63.5, Zn = 65.4, O = 16, S = 32, H = 1, Na = 23, N = 14, Cl = 35.5)

4) During the group-III cations analysis **precipitate (P)** is obtained after the addition of dilute HNO_3 and group reagents of IIIrd group cations.

precipitate (P) dissolve in hydrochloric acid and resulting solution is treated with potassium ferrocyanide solution, a blue **precipitate (Q)** is formed. If potassium ferrocyanide is added in excess then a blue product of **composition (R)** is formed. This **(R)** tends to form a colloidal solution and cannot be filtered.

Find **n₁ + n₂ + n₃**

where,

Oxidation Number of Metal in **(P)** = **n₁**

Number of Carbon in **(Q)** = **n₂**

Number of iron in **(R)** = **n₃**

5) An element 'E' of 4th period of periodic table which exist in B.C.C. lattice structure, have high

melting point and large heat of atomisation. Element 'E' have E^{+2}/E electrode potential = -1.18 V and E^{+3}/E^{+2} electrode potential = -0.26 V.

Element 'E' produces violet coloured aquated ion $[E(H_2O)_6]^{+2}$, which can reduce H_2O ($1M H^+$) with the liberation of H_2 , thus the violet solution soon turns green through oxidation to another aquated paramagnetic ion.

If atomic number of E = 'x' and highest oxidation state of E = '+y' then **find (x + y)**

6) Hexadentate ligand forms very stable complexes (usually octahedral structures) with most of the transition metals.

Find **($n_1 + n_2$)**

where,

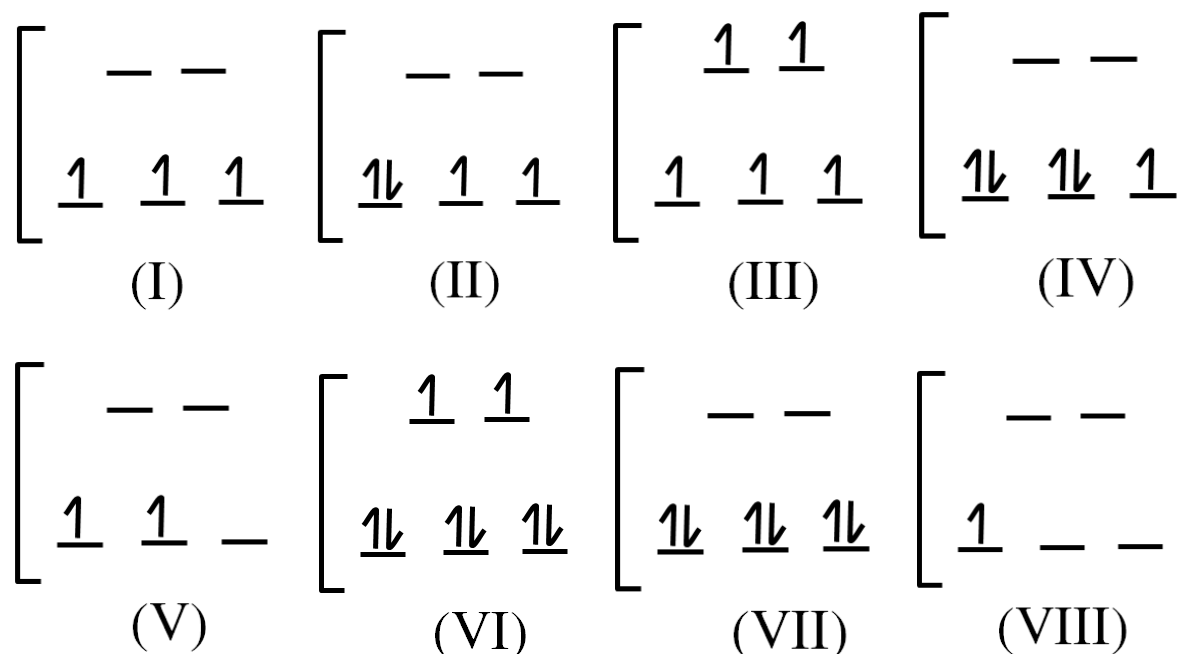
n_1 = Number of five membered chelate ring(s) which must involve O-Co-N bond angle in $[Co(EDTA)]^{3-}$.

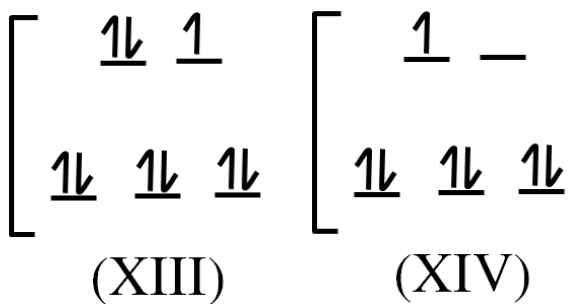
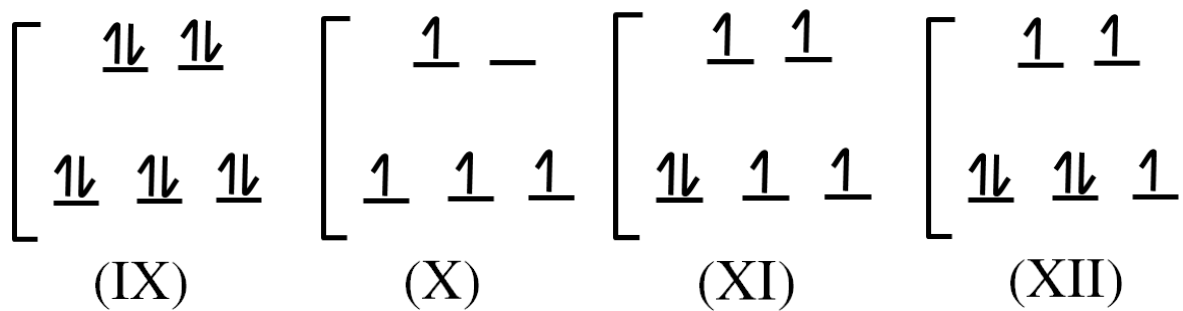
n_2 = Number of cis O-Co-O bond angle(s) in $[Co(EDTA)]^{3-}$
(where $EDTA^{4-}$ is ethylenediamine tetraacetate ion)

7) Number of species that do not exist among the following.

VOF_3 , CuI_2 , MnF_7 , Cr_2O_3 , Cu_2I_2 , $ZnCl_2$, FeI_3 , $FeBr_2$, MnO_2 , $CrO(O_2)_2$, Sc^{+4} , K_3CrO_8

8) As per CFT consider the following different spin state according to ligand field strength.
(In octahedral complexes)





(P) = Number of spin state which is/are can be obtained by irrespective nature of ligand.

(Q) = Number of spin state which can be obtained only when CFSE > PE but not in other cases.

Find sum of **P** & **Q**.

ANSWER KEYS

PART-1 : MATHEMATICS

SECTION-I (i)

Q.	1	2	3	4	5	6
A.	A,B,D	B,D	A,D	B,D	B,D	A,B,C

SECTION-I (ii)

Q.	7	8	9	10
A.	A	C	C	A

SECTION-II

Q.	11	12	13	14	15	16	17	18
A.	4.00	10.00	5.00	141.00	152.00	4.00	5.00	140.00

PART-2 : PHYSICS

SECTION-I (i)

Q.	19	20	21	22	23	24
A.	B,C	A,B,C	A,C	B,C	A,B,C	A,B,D

SECTION-I (ii)

Q.	25	26	27	28
A.	A	C	A	A

SECTION-II

Q.	29	30	31	32	33	34	35	36
A.	2.25	0.55 to 0.56	2.86 to 2.89	0.56 to 0.58	3.80 to 3.82	25.20	2.33	3.36

PART-3 : CHEMISTRY

SECTION-I (i)

Q.	37	38	39	40	41	42
A.	A,C,D	A,D	B,D	B,C	B,D	B,C

SECTION-I (ii)

Q.	43	44	45	46
A.	B	B	C	B

SECTION-II

Q.	47	48	49	50	51	52	53	54
A.	4.00	18.00	7.73 to 7.75	23.00	28.00	9.00	4.00	10.00

SOLUTIONS

PART-1 : MATHEMATICS

$$1) n_1 = 10 \times 10 \times 10 = 10^3$$

$$n_2 = {}^8C_2 + 2 {}^8C_1 = 44$$

$$n_3 = {}^{10}C_4 = 210$$

$$n_4 = {}^{10}P_4 = 5040$$

$$2) \sin^{-1}(e^x) = \sin^{-1}(x^2)$$

for $\sin^{-1}(e^x)$;

$$e^x \in (0, 1] \Rightarrow x \in (-\infty, 0] \quad \dots(1)$$

for $\sin^{-1}(e^2)$;

$$x^2 \in [0, 1] \Rightarrow x \in [-1, 1] \quad \dots(2)$$

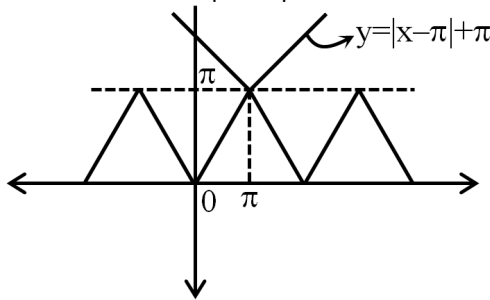
from (1) and (2)

$$x \in [-1, 0]$$

$$\Rightarrow \boxed{\alpha < 0}$$

$$\cos^{-1}\left(\frac{1 - \tan^2\left(\frac{x}{2}\right)}{1 + \tan^2\left(\frac{x}{2}\right)}\right) = |x - \pi| + \pi$$

$$\Rightarrow \cos^{-1}(\cos x) = |x - \pi| + \pi, \quad x \neq (2n + 1)\pi$$



But $x \neq \pi$

No solution

$$\boxed{\beta = 0}$$

$$3) \sin^{-1}\sin 10 = 3\pi - 10, \tan^{-1}\tan(-6) = 2\pi - 6, \cos^{-1}\cos 12 = 4\pi - 12$$

$$\cot^{-1}\cot 4 = 4 - \pi, \sec^{-1}\sec 9 = 9 - 2\pi, \operatorname{cosec}^{-1}\operatorname{cosec} 7 = 7 - 2\pi$$

$$\Rightarrow \text{sum} = 8\pi - 28$$

$$4) f_1(x) = x^2 + 4x + 2 = (x + 2)^2 - 2$$

$$f_2(x) = ((x + 2)^2 - 2 + 2)^2 - 2 = ((x + 2)^2)^2 - 2$$

$$f_3(x) = (f_1(x) + 2)^{2 \times 2} - 2 = [(x + 2)^2 - 2 + 2]^{2 \times 2} - 2$$

$$f_3(x) = (x + 2)^{2 \times 2 \times 2} - 2$$

$\vdots$

$$f_n(x) = (x + 2)^{\underbrace{2 \times 2 \times 2 \times \dots \times 2}_{n \text{ times}}} - 2$$

$$\therefore f_{2025}(x) = (x + 2)^{2^{2025}} - 2$$

$$\text{Let } n = 2^{2025} = \text{even}$$

$$(x + 2)^{2^{2025}} - 2 = (x + 2)^n - 2$$

$$\begin{aligned}
&= \{ {}^nC_0 x^n + {}^nC_1 x^{n-1} \cdot 2^1 + {}^nC_2 x^{n-2} \cdot 2^2 + \dots \} \\
(x-2)^n - 2 &= {}^nC_0 x^n - {}^nC_1 x^{n-1} 2^1 + {}^nC_2 x^{n-2} 2^2 - \dots \\
&\text{add for even power of } x \\
&= {}^nC_0 x^n + {}^nC_2 x^{n-2} \cdot 2^2 + {}^nC_4 x^{n-4} \cdot 2^4 + \dots - 2 \\
&= \frac{(x+2)^n - 2 + (x-2)^n - 2}{2}
\end{aligned}$$

put $x = 1$ to get sum of coefficient of even power of x .

$$\begin{aligned}
\therefore \text{required sum} &= \frac{(1+2)^{2^n} - 2 + (1-2)^{2^n} - 2}{2} \\
&= \frac{3^{2^n} - 4 + (-1)^{2^n}}{2} \\
&= \frac{3^{2^n} - 3}{2} \\
\therefore S_n &= \frac{3^{2^n} - 3}{2}, \text{ now verify options}
\end{aligned}$$

5) **(A)** Say $A = \{0, 1, 2\}$, we say $R_2 = \{(0, 0), (1, 1), (2, 2), (0, 1), (1, 0)\}$ and

$R_1 = \{(0, 0), (1, 1), (2, 2), (1, 2), (2, 1)\}$.

Then $R_1 \cdot R_2 = \{(0, 0), (1, 1), (1, 2), (2, 2), (2, 1), (0, 1), (0, 2), (1, 0)\}$

(B,C) Number of reflexive relations is 2^{n^2-n} and number of symmetric relations is $2^{\frac{n(n+1)}{2}}$

(D) Corresponding to every partition of the set $\{1, 2, 3, 4\}$, there exists a unique equivalence relation. So we count every type of unordered partitions of the set of 4 elements into one block, two block, three block and four block partitions, as shown below :

$$\begin{array}{c}
4 \\
\downarrow \\
4 = \frac{4!}{4!} = 1
\end{array}$$

$$\begin{array}{c}
4 \\
\swarrow \quad \searrow \\
1 \quad 3 \\
2 \quad 2
\end{array}
= \frac{4!}{3!} + \frac{4!}{2!2!2!} = 4 + 3 = 7$$

$$\begin{array}{c}
4 \\
\swarrow \quad \downarrow \quad \searrow \\
2 \quad 1 \quad 1
\end{array}$$

$$\begin{array}{c}
4 \\
\swarrow \quad \downarrow \quad \searrow \quad \searrow \\
1 \quad 1 \quad 1 \quad 1
\end{array}
= \frac{4!}{4!} = 1$$

Total = $1 + 7 + 6 + 1 = 15$

So the number of equivalence relations on the set $\{1, 2, 3, 4\}$ is 15.

$$\begin{aligned}
6) \phi(x) &= \tan \left(\frac{\cos^{-1}(\sin(\cos^{-1}x)) + \sin^{-1}(\cos(\frac{\pi}{2} - \cos^{-1}x))}{2} \right) \\
&= \tan \left(\frac{\cos^{-1}(\sin(\cos^{-1}x)) + \sin^{-1}(\sin(\cos^{-1}x))}{2} \right)
\end{aligned}$$

$$= \tan\left(\frac{\pi/2}{2}\right) = \tan\left(\frac{\pi}{4}\right) = 1$$

7) (I) We have $f(x) = \sqrt[5]{x} + \sin^{-1}x$

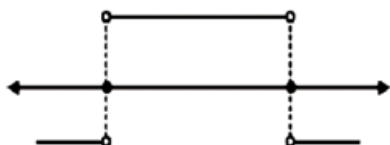
Clearly domain of $f(x) = [-1, 1]$.

Also, $f(x)$ is increasing so $f(x)$ is one-one function

(II) $f(x) = \operatorname{sgn} \frac{(1 - |x|)}{(1 + |x|)}$

$$D_f = \mathbb{R}$$

$$R_f = \{-1, 0, 1\} \text{ even function}$$



(III) $f(x) = \frac{2^{[-x]}}{2^{\{x\}}} - 2^{|x|} = 2^{-x} - 2^{-x} = 0 \quad \forall x \leq 0$

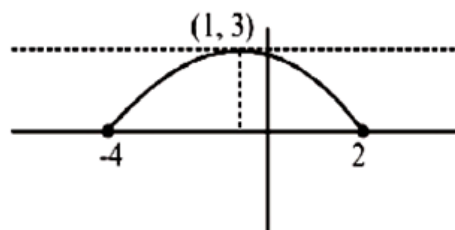
(IV) For domain of $f(x)$ we must have $8 - 2x - x^2 \geq 0$

$$\Rightarrow x^2 + 2x - 8 \leq 0$$

$$\Rightarrow (x + 4)(x - 2) \leq 0$$

$$\Rightarrow x \in [-4, 2]$$

$$\Rightarrow R_f = [0, 3]$$



8) (I) $D_f : x \in \left(-\infty, -\frac{1}{4}\right) - \left\{-\frac{3}{4}\right\}$

(II) $x^2 + 4x \geq 2x^2 + 3$

$$x^2 - 4x + 3 \leq 0$$

$$x \in [1, 3]$$

$$D_f : x \in \{1, 2, 3\}$$

$$f(1) = 1; f(2) = \sqrt{12}; f(3) = 1$$

$\Rightarrow$ only one integer in the range.

(III) $\{f(x)\} = 0 \Rightarrow f(x) = (3 - x^7)^{1/7}$

$$f(f(x)) = x$$

(IV) $2\tan^{-1}(\sec^2 \pi x) \geq \frac{\pi}{2}$

$$\Rightarrow \sin^{-1}(x^3 - x^2 + x + 2) \leq \frac{\pi}{2}$$

$$\square x^3 - x^2 + x + 1 = 0$$

No integral solution

$$9) (I) f(n) = \sum_{k=-n}^{-1} \left(\cos^{-1} \left(\frac{1}{k} \right) - \tan^{-1}(k) \right) + \sum_{k=1}^n \left(\cos^{-1} \frac{1}{k} - \tan^{-1} k \right)$$

$$= \sum_{k=-n}^{-1} \left(\tan^{-1} k + \pi - \tan^{-1} k \right) + 0$$

$$\sum_{n=2}^{10} n\pi + (n-1)\pi = \sum_{n=2}^{10} (2n-1)\pi = 99\pi$$

$$(II) \text{ Let } k = \sum_{n=1}^{20} \sum_{m=1}^{20} \tan^{-1} \left(\frac{m}{n} \right); k = \sum_{n=1}^{20} \sum_{m=1}^{20} \tan^{-1} \left(\frac{n}{m} \right)$$

$$\Rightarrow 2k = 400 \frac{\pi}{2} \Rightarrow k = 100\pi \Rightarrow \sum_{p=1}^3 \sum_{r=1}^p 100\pi = \sum_{p=1}^3 100\pi p = 100\pi \times \frac{3 \times 4}{2}$$

$$= 600\pi$$

$$(III) ([x] + [y] + [z])([x] - [y])^2 + ([y] - [z])^2 + ([z] - [x])^2 = 8$$

possible only when $[x] + [y] + [z] = 4$

$$[x] = 2, [y] = 1, [z] = 1$$

$$\text{or } [x] = 1, [y] = 2, [z] = 1$$

$$[x] = 1, [y] = 1, [z] = 2$$

$$(IV) 2\cos^{-1}(1-x) = \pi + 3\cos^{-1}x$$

This is possible only when

$$3\cos^{-1}x = 0 \text{ \& } \cos^{-1}(1-x) = \frac{\pi}{2}$$

$$10) (I) \text{ Number of asymmetric relation} = 3^{\frac{n^2-n}{2}}$$

$$= 3^{325} = p^q$$

$$\Rightarrow p + q = 328$$

$$(II) n(A \times A) = 26 \times 26$$

For reflexive relation (a, a) must be taken hence rest of the selection will be from $26 \times 26 - 26$
 $= 650$ elements of $A \times A$

$\Rightarrow$ Total number of reflexive relations are 2^{650}

Total number of unordered pairs (a, b), $a \neq b$ are ${}^{26}C_2 = 325$.

For each such pair, include both or exclude both $= 2$ choices each

So, reflexive + symmetric relations are 2^{325}

Hence reflexive but not symmetric are

$$2^{650} - 2^{325} = 2^{325} (2^{325} - 1)$$

(III) Total number of unordered pairs (a, b), $a \neq b$ are ${}^{26}C_2 = 325$.

Total ways of choosing (a, b) & (b, a) simultaneously but not all (a, a) are

$$\left({}^{325}C_0 + {}^{325}C_1 + {}^{325}C_2 + \dots + {}^{325}C_{325} \right)$$

$$\left({}^{26}C_0 + {}^{26}C_1 + {}^{26}C_2 + \dots + {}^{26}C_{25} \right) - 1$$

$$= 2^{325} (2^{26} - 1) - 1$$

(IV) Total non-empty relations are $2^{676} - 1$

Symmetric relation are $2^{325} \cdot 2^{26} = 2^{351}$

Reflexive relations are 2^{650}

Symmetric and reflexive relations are 2^{325}

$\Rightarrow$ Neither symmetric nor reflexive are $2^{676} - 2^{650} - 2^{351} + 2^{325} - 1 + 1$

$$= 2^{325} (2^{325} - 1) (2^{26} - 1)$$

$$\begin{aligned}
 11) \quad f^3(a) &= a + \frac{(a+1)\sqrt{8a-1}}{\sqrt{27}} + a - \frac{(a+1)\sqrt{8a-1}}{\sqrt{27}} \\
 &+ 3 \left(a + \frac{(a+1)\sqrt{8a-1}}{\sqrt{27}} \right) \left(a - \frac{(a+1)\sqrt{8a-1}}{\sqrt{27}} \right) f(a) \\
 \Rightarrow f^3(a) &= 2a + (1-2a)f(a) \Rightarrow f(a) = 1
 \end{aligned}$$

$$\begin{aligned}
 12) \quad \sin^{-1} \left(\frac{2x}{1+x^2} \right) &= \begin{cases} -\pi - 2\tan^{-1}x, & x < -1 \\ 2\tan^{-1}x, & -1 \leq x \leq 1 \\ \pi - 2\tan^{-1}x, & x > 1 \end{cases} \\
 \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) &= \begin{cases} -2\tan^{-1}x, & x < 0 \\ 2\tan^{-1}x, & x > 0 \end{cases} \\
 \tan^{-1} \left(\frac{2x}{1-x^2} \right) &= \begin{cases} 2\tan^{-1}x, & x < -1 \\ 2\tan^{-1}x, & -1 < x < 1 \\ 2\tan^{-1}x - \pi, & x > 1 \end{cases} \\
 f(x) &= \begin{cases} -2\tan^{-1}x, & x < -1 \\ 2\tan^{-1}x, & -1 < x < 0 \\ 6\tan^{-1}x, & 0 < x < 1 \\ 2\tan^{-1}x, & x > 1 \end{cases}
 \end{aligned}$$

$$13) \quad f(x) = \sqrt{(2x+9) - \left(\frac{2x}{1-\sqrt{1+2x}} \right)^2}$$

$$\square x \neq 0 \text{ \& } 1+2x \geq 0 \Rightarrow x \geq -\frac{1}{2}$$

$$(2x+9) - \left(\frac{2x}{1-\sqrt{2x+1}} \right)^2 \geq 0$$

$$(t^2+8) - \left(\frac{t^2-1}{1-t} \right)^2 \geq 0$$

$$(t^2+8) - (-t-1)^2 \geq 0$$

$$t^2 + 8 - (t^2 + 1 + 2t) \geq 0$$

$$7 - 2t \geq 0$$

$$t \leq \frac{7}{2} \quad \square \quad t \in \left[0, \frac{7}{3} \right]$$

$$\text{Let } \sqrt{2x+1} = t \quad t \geq 0$$

$$\sqrt{2x+1} \in \left[0, \frac{7}{3} \right]$$

$$2x+1 \in \left[0, \frac{49}{9} \right]$$

$$x \in \left[-\frac{1}{2}, \frac{45}{18} \right]$$

$$D_f: \left[-\frac{1}{2}, \frac{45}{8}\right] - \{0\}$$

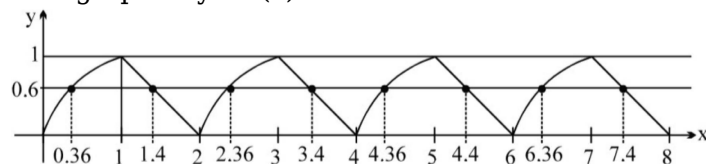
Integers in its domain = 1, 2, 3, 4, 5

$$\begin{aligned}
 14) \quad \alpha &= \lim_{r \rightarrow \infty} \sum_{n=1}^r \tan^{-1} \left\{ \frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16} \right\} \\
 &= \lim_{r \rightarrow \infty} \sum_{n=1}^r \tan^{-1} \left\{ \frac{2\left(n + \frac{1}{2}\right)}{1 + \left(n + \frac{1}{2}\right)^4 - \left(n + \frac{1}{2}\right)^2} \right\} \\
 &= \lim_{r \rightarrow \infty} \sum_{n=1}^r \tan^{-1} \left\{ \frac{\left\{ \left(n + \frac{1}{2}\right)^2 + \left(n + \frac{1}{2}\right) \right\} - \left\{ \left(n + \frac{1}{2}\right)^2 - \left(n + \frac{1}{2}\right) \right\}}{1 + \left\{ \left(n + \frac{1}{2}\right)^2 + \left(n + \frac{1}{2}\right) \right\} \left\{ \left(n + \frac{1}{2}\right)^2 - \left(n + \frac{1}{2}\right) \right\}} \right\} \\
 &= \lim_{r \rightarrow \infty} \left[\sum_{n=1}^r \tan^{-1} \left\{ \left(n + \frac{1}{2}\right)^2 + \left(n + \frac{1}{2}\right) \right\} - \tan^{-1} \left\{ \left(n + \frac{1}{2}\right)^2 - \left(n + \frac{1}{2}\right) \right\} \right] \\
 &= \lim_{r \rightarrow \infty} \tan^{-1} \left\{ \left(r + \frac{1}{2}\right)^2 + \left(r + \frac{1}{2}\right) \right\} - \tan^{-1} \frac{3}{4} \\
 &= \frac{\pi}{2} - \tan^{-1} \frac{3}{4} = \cot^{-1} \frac{3}{4} \Rightarrow \cot \alpha = \frac{3}{4} \\
 &\Rightarrow \sin \alpha = \frac{4}{5} \\
 p^2 + q^3 &= 16 + 125 = 141
 \end{aligned}$$

15) Since $f(x+2) = f(x)$

$\Rightarrow f$ is periodic with period = 2 $\Rightarrow$ period of $g = 2/3$

now graph of $y = f(x)$



$$f(x) = 0.6 \Rightarrow \ln(0, 1) \quad \boxed{x = 0.36}$$

$$\ln(1, 2) \quad y = 2 - x \Rightarrow 2 - 0.6 = 1.4$$

$$\ln(3, 4) \quad x = 3.4$$

$$\ln(4, 5) \quad x = 4.36$$

$$\ln(5, 6) \quad x = 5.4$$

$$\ln(6, 7) \quad x = 6.36$$

$$\square \quad M = 6.36 + 4.36 + 5.4 + 3.4 = 19.52 \Rightarrow [M] = 19$$

now $\mathbf{N}$: period of $g(x) = \frac{2}{3}$

$$g(x) = 4f(3x) + 1$$

$$\mathbf{P} : g'(x) = 12f'(3x)$$

$$g'(6.75) = 12 \cdot f'(3 \times 6.75) = 12f'(20.25)$$

since f has a period '2'

$\square f'(x)$ also has a period '2'

$$f'(20 + 0.25) = f'(0.25) = f'(1/4)$$

$$\square g'(6.75) = 12f'(1/4)$$

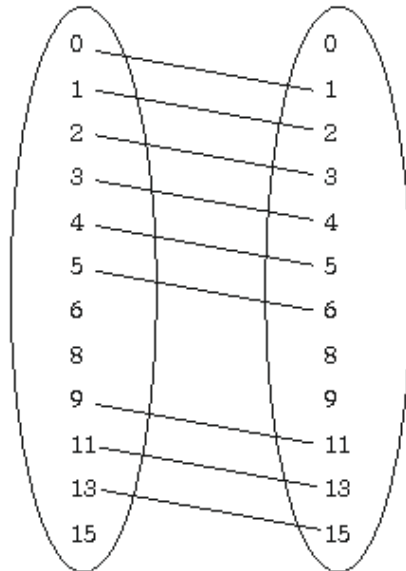
$$f(x) = \sqrt{x} \text{ in } (0, 1)$$

$$f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'\left(\frac{1}{4}\right) = 1$$

$$\Rightarrow g'(6.75) = 12$$

$$\square [M]NP = 19 \times \frac{2}{3} \times 12 = 19 \times 8 = 152$$

$$16) A = \{0, 1, 2, 3, 4, 5, 6, 8, 9, 11, 13, 15\}$$



$$17) \text{ Let } g(x) = \frac{x^3 - 3x + 6}{2}, g'(x) = 0 \Rightarrow x = \pm 1$$

$$\& g(1) = 2 \& g(-1) = 4$$

$$\square y = 2, y = 4 \text{ touch } y = g(x) \text{ at } x = 1 \& x = -1 \text{ respectively}$$

$$\& g(x) = 2 \Rightarrow x^3 - 3x + 2 = 0 \Rightarrow (x - 1)^2(x - 2) = 0$$

$$\& g(x) = 4 \Rightarrow x^3 - 3x - 2 = 0 \Rightarrow (x + 1)^2(x - 2) = 0$$

$$\& g(x) = 2 \Rightarrow x = 1, 1, -2$$

$$\& g(x) = 4 \Rightarrow x = -1, -1, 2.$$

Now due to $(x - \alpha)$ a void, will be

Created in $y = g(x)$, for the given

Condition to be true that void can be created

$$\forall x \in (-\infty, -2) \cup (2, \infty)$$

$$\therefore \alpha \in \mathbb{R} - [-2, 2]$$

$$18) D_f : x \in [-1, 1]$$

$$f(x) = \frac{1}{\pi} \left(\frac{\pi}{2} + \tan^{-1} x \right) + \frac{(x+1)}{(x+1)^2 + 9}$$

$$M = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{4} \right) + \frac{2}{4+9}$$

$$= \frac{3}{4} + \frac{2}{13} = \frac{47}{52}$$

19)

Correct Answer is (B,C)

20)

Correct Answer is (A,B,C)

21)

Correct Answer is (A,C)

22)

Correct Answer is (B,C)

23)

Correct Answer is (A,B,C)

24)

Correct Answer is (A,B,D)

25)

Correct Answer is (A)

26)

Correct Answer is (C)

27)

Correct Answer is (A)

28)

Correct Answer is (A)

29)

Correct Answer is (2.25)

30)

Correct Answer is (0.55 to 0.56)

31)

Correct Answer is (2.86 to 2.89)

32)

Correct Answer is (0.56 to 0.58)

33)

Correct Answer is (3.80 to 3.82)

34)

Correct Answer is (25.20)

35)

Correct Answer is (2.33)

36)

Correct Answer is (3.36)

PART-3 : CHEMISTRY

37)

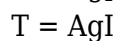
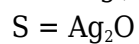
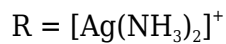
The correct answer is option (A), (C), (D)

38)

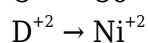
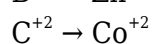
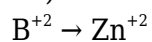
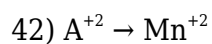
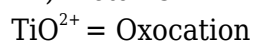
The correct answer is option (A), (D)

- 39) (A) Redox reaction between Mn^{2+} & BiO_3^-
(B) Coexist together
(C) Precipitate of $\text{Al}(\text{OH})_3$
(D) Coexist together

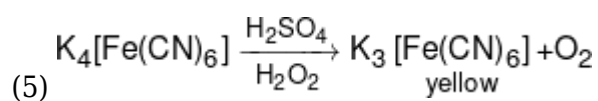
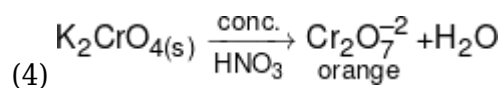
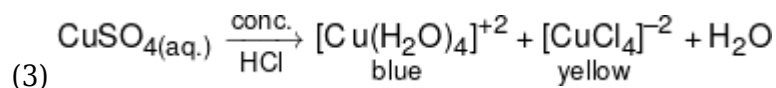
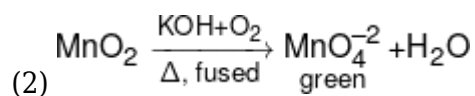
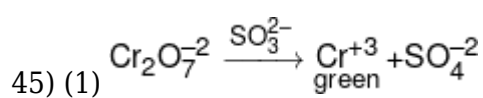
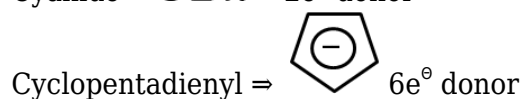
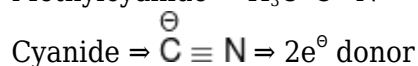
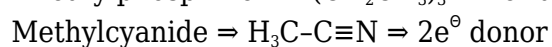
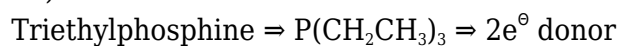
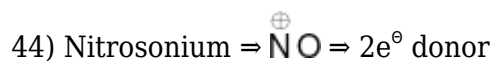
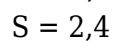
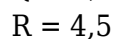
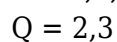
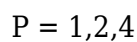
- 40) P = AgNO_3
Q = AgCl



41) Metal is Ti



43)



46) P = 4,5

$$Q = 2$$

$$R = 3$$

$$S = 1$$

47) (ii), (iii), (iv), (vi)

$$48) x = 4$$

$$y = 3$$

$$z = 2$$

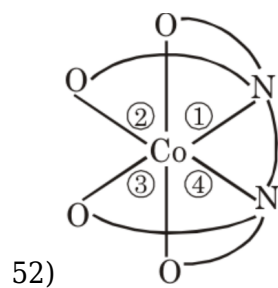
$$t = 2$$

49)

The correct answer is 7.73-7.75

50) P : $\text{Fe}(\text{OH})_3$
 Q : $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 R : $\text{KFe}[\text{Fe}(\text{CN})_6]$

51) Atomic number $\Rightarrow$ 23 (Vanadium)
 X = 23
 Y = +5



53) CuI_2 , MnF_7 , FeI_3 , Sc^{+4}

54) P = 6 (I, V, VI, VIII, IX, XIII)
 Q = 4 (II, IV, VII, XIV)